DATA MINING & MACHINE LEARNING

HEART: Heartbeat Extraction and Arrhythmia Recognition Tool

Data Mining & Machine Learning Project
M.Sc. in Artificial Intelligence and Data Engineering — Academic Year 2025–2026

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OVERVIEW

Presentation Outline

01 **Problem & Dataset**
Task definition · features · EDA

02 **Pipeline & Experiments**
Preprocessing · model selection · hyperparameter optimization

PART 1

03 **Evaluation & Results**
Metrics · cross-validation · final model performance

04 **Explainability**
Global & local techniques · consistency with domain knowledge

PART 2

05 **Critical Discussion**
Error analysis · limitations · future directions

PART 1 — TECHNICAL

Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

PROBLEM

Problem Definition & Dataset

The Problem

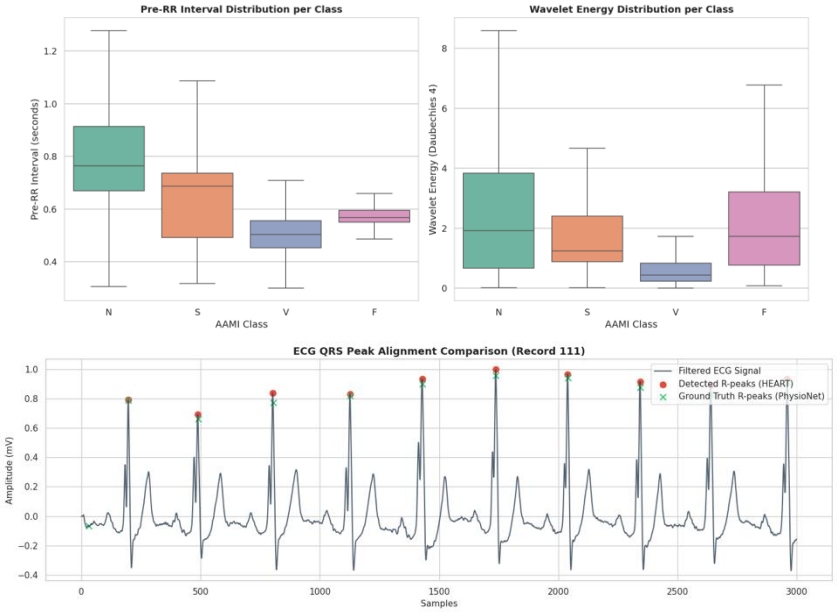
- **Arrhythmias** are abnormalities in the heart rhythm that can indicate serious cardiovascular conditions. We aim to automatically **classify ECG heartbeats** into clinically relevant categories using machine learning techniques.
- Task typer: **Supervised Multi-Class Classification** (four AAMI heartbeat classes: N, S, V, F).
- Automated arrhythmia detection can **support cardiologists** by reducing the workload of manual ECG inspection and enabling earlier diagnosis.
- The problem is challenging because **ECG classes are highly imbalanced** and some arrhythmias exhibit heartbeat morphologies very similar to normal beats.

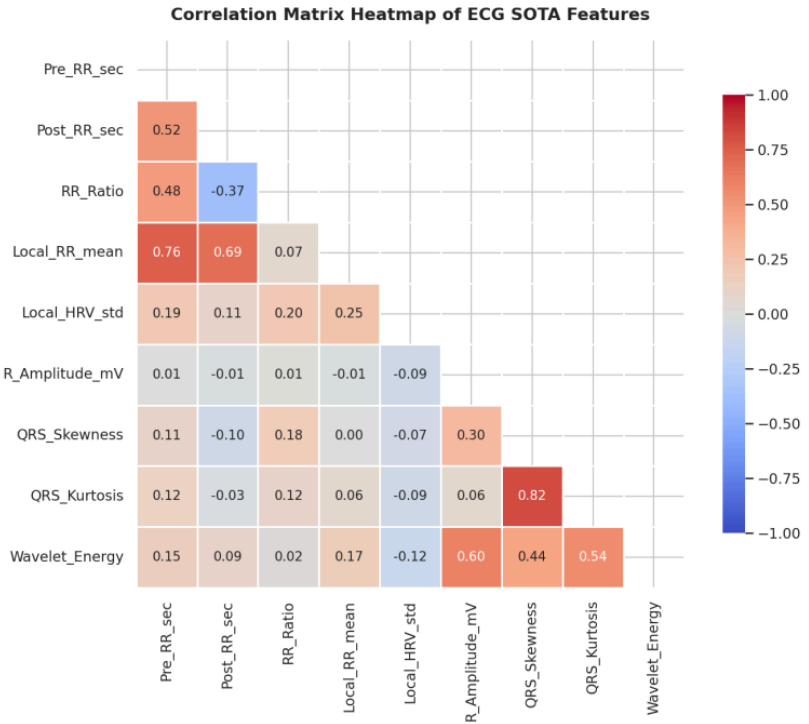
Dataset at a Glance

Task	*Multi-class classification*
Dataset	*MIT-BIH Arrhythmia Database, 99.039 beats*
Target	*F: Fusion, N: Normal, S: Supraventricular, V: Ventricular*
Features	*5 temporal rhythm, 3 morphology, 1 wavelet energy*
Missing values	*0.0%*
Class balance	*N: 99.64%, V: 5.67%, S: 2.67%, F: 0.79% (imbalanced)*
Split	*Inter-Patient (De Chazal Split) – Train 49,5%, Test: 50,5%*
Metric(s)	*F_1, F_2, ROC-AUC*

PROBLEM

Exploratory Data Analysis





Key findings from EDA:

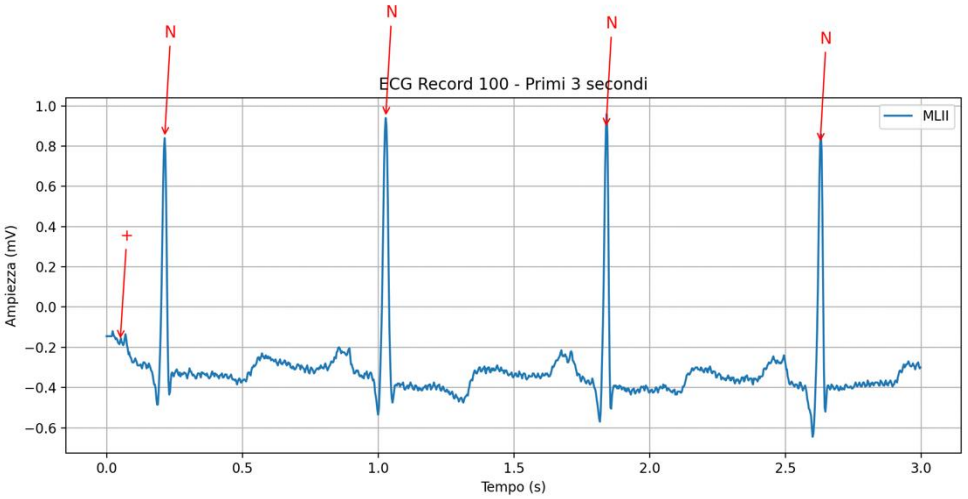
- The dataset is highly **imbalanced**: normal beats represent 88.6% of all observations.
- Supraventricular and Ventricular beats show shorted Pre_RR intervals, indicating premature contractions.
- Morphological and temporal features are weakly correlated, suggesting complementary diagnostic information.
- Missing values: 0%.

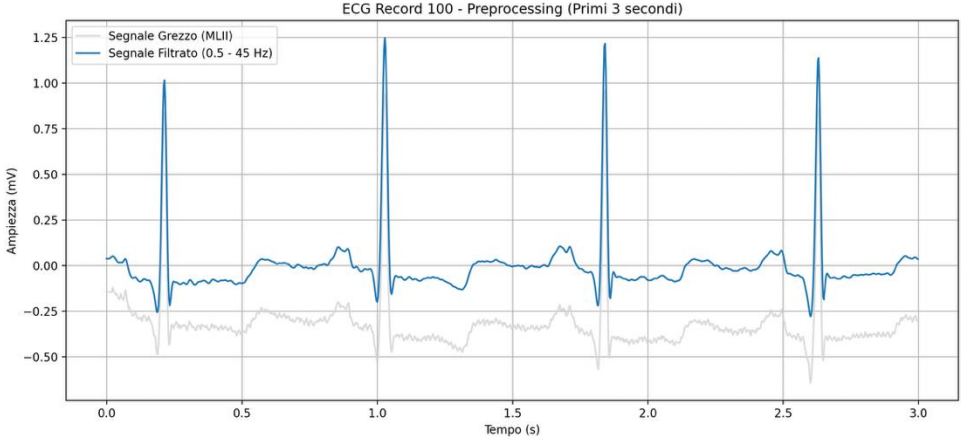
PREPROCESSING

Preprocessing

Signal Cleaning:

- Raw ECG signals were filtered using a zero-phase 3rd order Butterworth bandpass filter (*0.5 – 45 Hz*)
- This removes baseline wander and high-frequency muscle noise while preserving the morphology of ECG waveforms





PIPELINE

Feature Engineering & Selection

Engineering

- The extraction of **nine physiological ECG features**, which are considered state-of-the-art from literature, has been performed.
- **Standardization** has been applied using the StandardScaler.
- **No categorical variables** are present in the dataset.
- **No imputation** is required, as there are no missing values (0%).

Selection Method

- Embedded/explainability-driven selection
- Permutation Feature Importance on SVM
- SHAP based analysis on RF/XGBoost
- Evalued on training data only
- All nine features retained

Why These Features?

- Time features capture **rhythm abnormalities**
- Morphological features capture QRS **shape distorsions**
- Wavelet energy captures **spectral information**
- Features were retained because they provide **complementary** clinical information

PIPELINE

Feature Engineering & Selection

Pre RR sec
Capture heartbeat prematurity

Local RR mean
Defines the patient's baseline heart rate

R Amptitude
Represents voltage changes

Post RR sec
Time between current and next R-peak

Local HRV std
Measure short-term heart rate variability

QRS Skewness
Asymmetry of the QRS complex

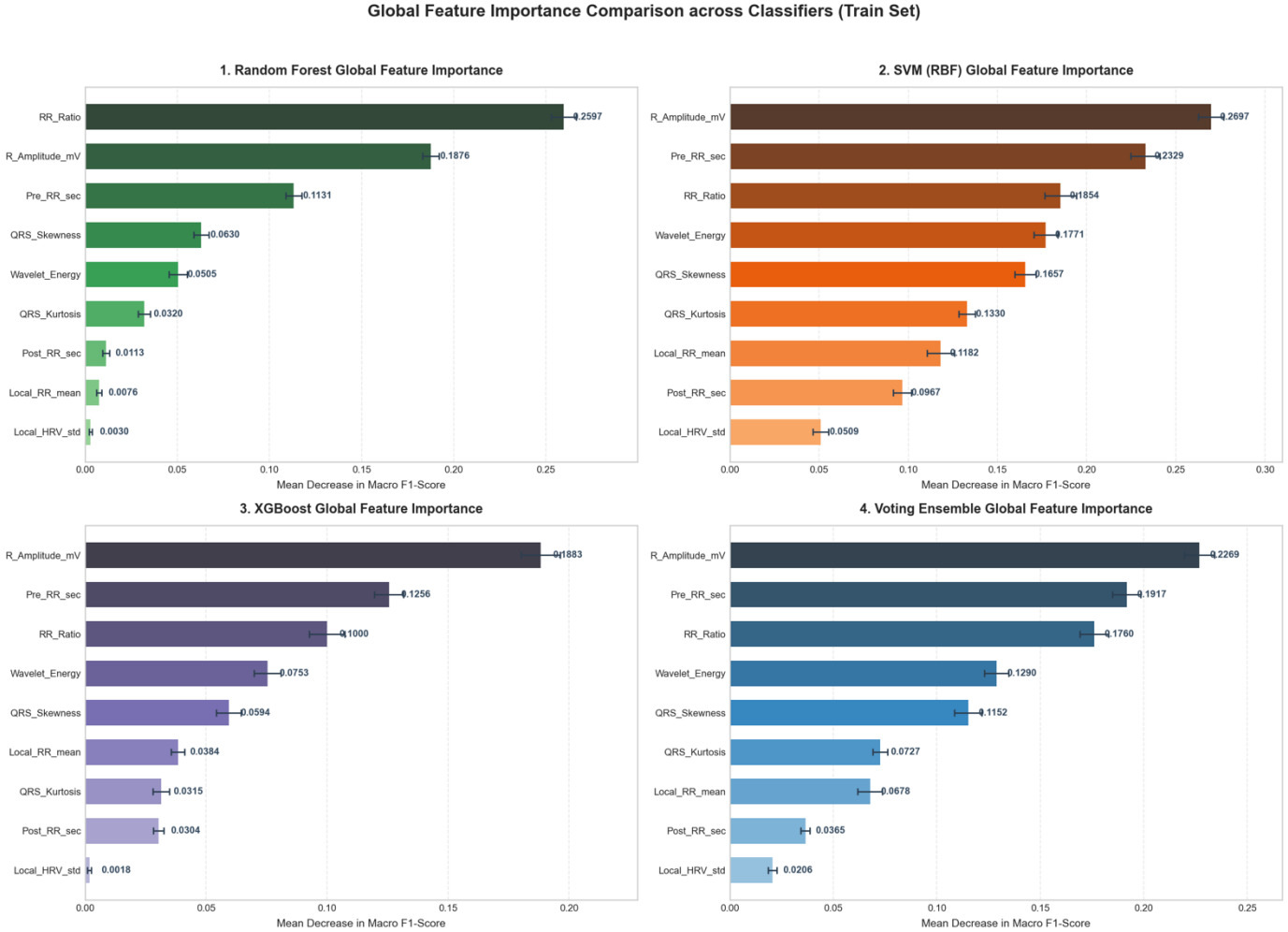
RR Ratio
Highlights sudden changes in cardiac rhythm

Wavelet Energy
Capture high-frequency QRS depolarization

QRS Kurtosis
Peakness of the QRS complex

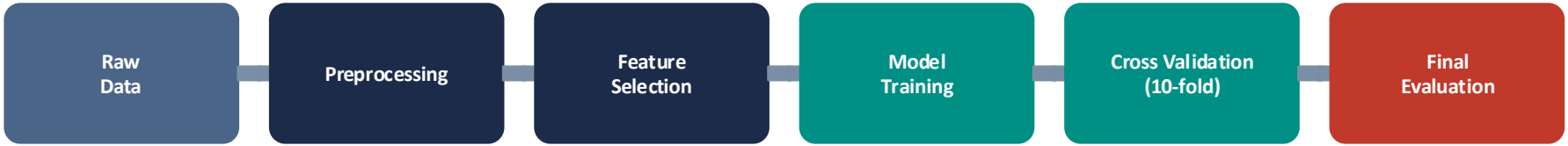
PIPELINE

Feature Engineering & Selection



PIPELINE

ML Pipeline: Avoiding Data Leakage & Bias



Data Leakage Prevention

- Inter-patient De Chazal split
- StandardScaler fitted only on training data
- Test patients never seen during training

Class Imbalance Handling

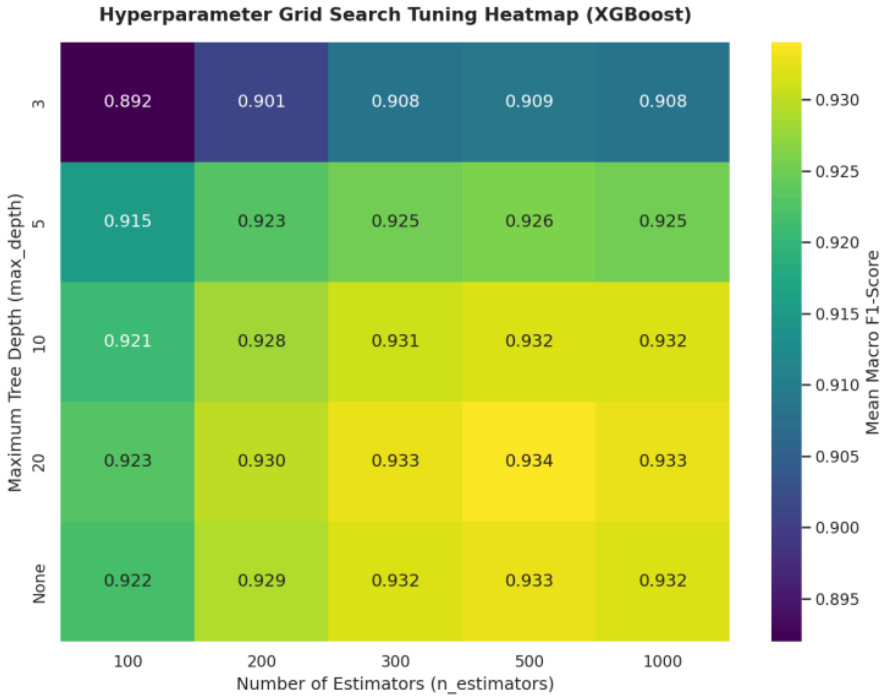
- Random Undersampling
- Applied only to training data
- Stratified Cross Validation

Metric Justification

- Primary metric: Macro F_2-score
- Recall is more important than precision in arrhythmia screening
- Missing an arrhythmia (FN) is more costly than generating an alarm (FP)
- Additional metrics: Macro F_1 and OvR ROC-AUC

PIPELINE

Model Selection & Hyperparameter Optimization



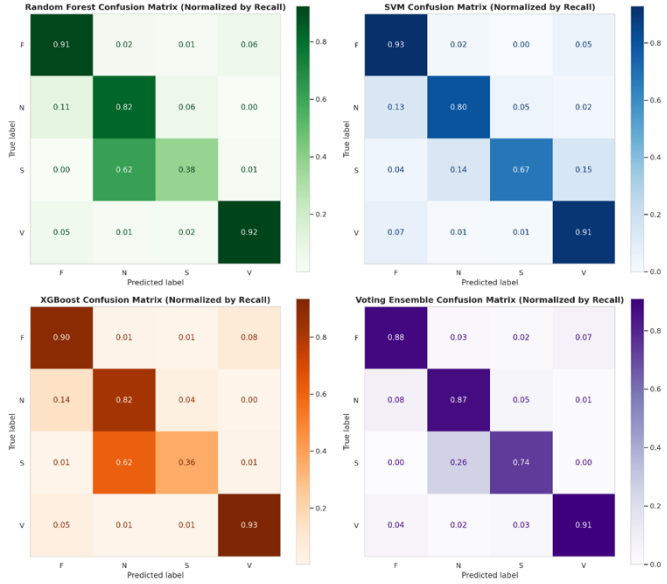
Example of HP tuning for XGBoost using Grid Search

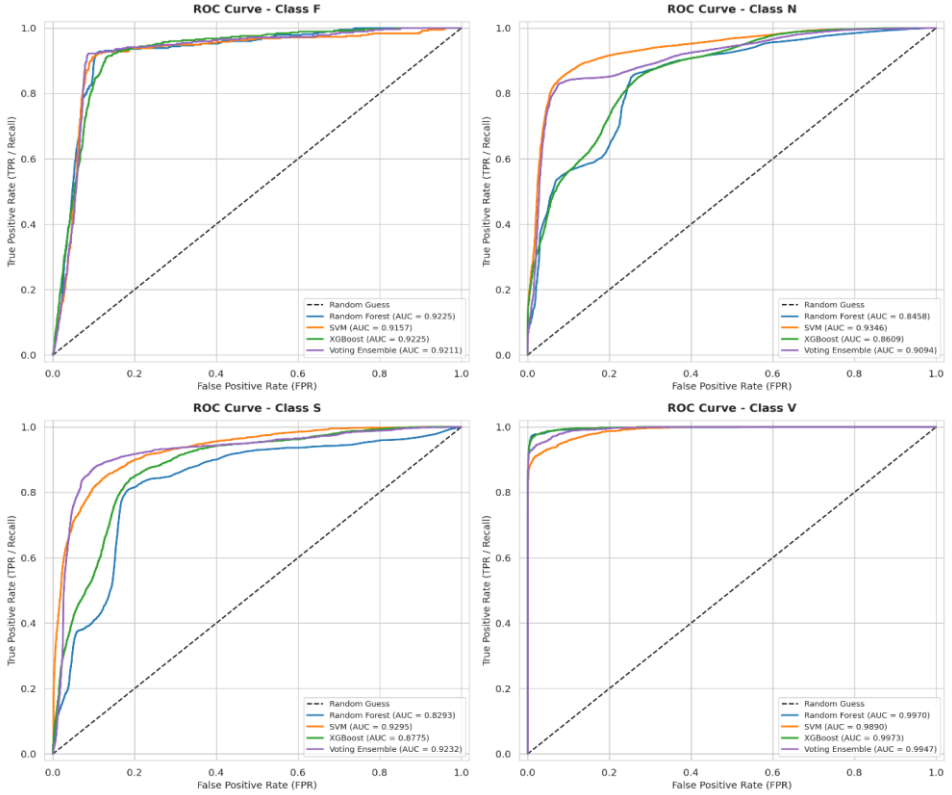
Key Decisions

Candidate models	*Random Forest, SVM, XGBoost*
CV strategy	*Stratified 10-fold CV*
Selection criterion	*Macro F1*
Chosen model	*Weighted Soft Voting Ensemble*
Search strategy	*Grid Search (RF, SVM) and Random Search (XGBoost)*
Search space	*RF: n_estimators = {100, 200, 500, 1000}, max_depth = {10, 20, None}, min_samples_split = {2, 5}; SVM: C = {0.1, 1, 10, 100}, n_estimators = {50, 100, 200, 500}; XGBoost: max_depth = {2, 3, 5, 7}, n_estimators = {50, 100, 200, 500}*
Best config	*RF: n_estimators = 1000, max_depth = 20, min_samples_split = 2; SVM: Kernel = rbf, C = 10, gamma = 'scale'; XGBoost: n_estimators = 500, max_depth = 3, learning_rate = 0.1*
Statistical test	*Wilcoxon signed-rank Test* *RF vs SVM: p = 0.019; XGB vs SVM: p = 0.039*

RESULTS

Test-Set Results





Key Metrics of the Voting Ensemble (on Test Set)

0.62	0.6825	0.9371	Majority Class
Macro F1	Macro F2	Macro AUC	N Class

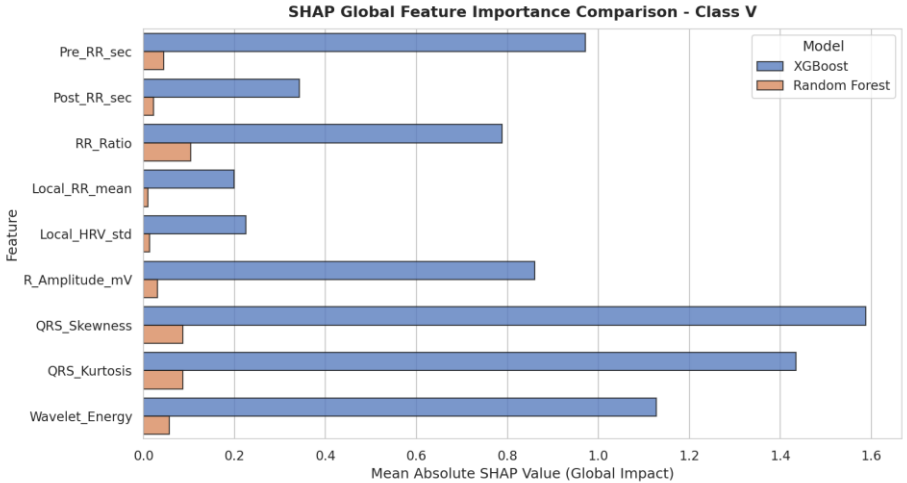
PART 2 — CRITICAL ANALYSIS

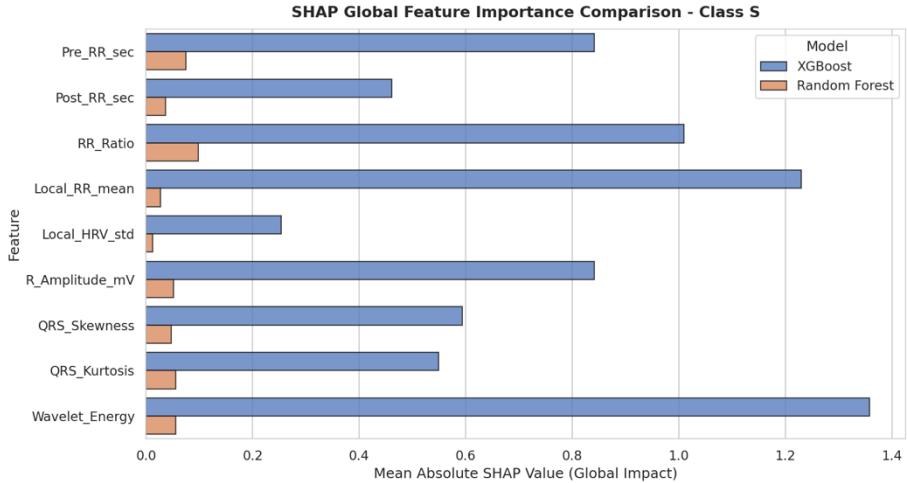
Explainability & Critical Discussion

How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

EXPLAINABILITY

Global Explainability: What Drives the Model Overall?



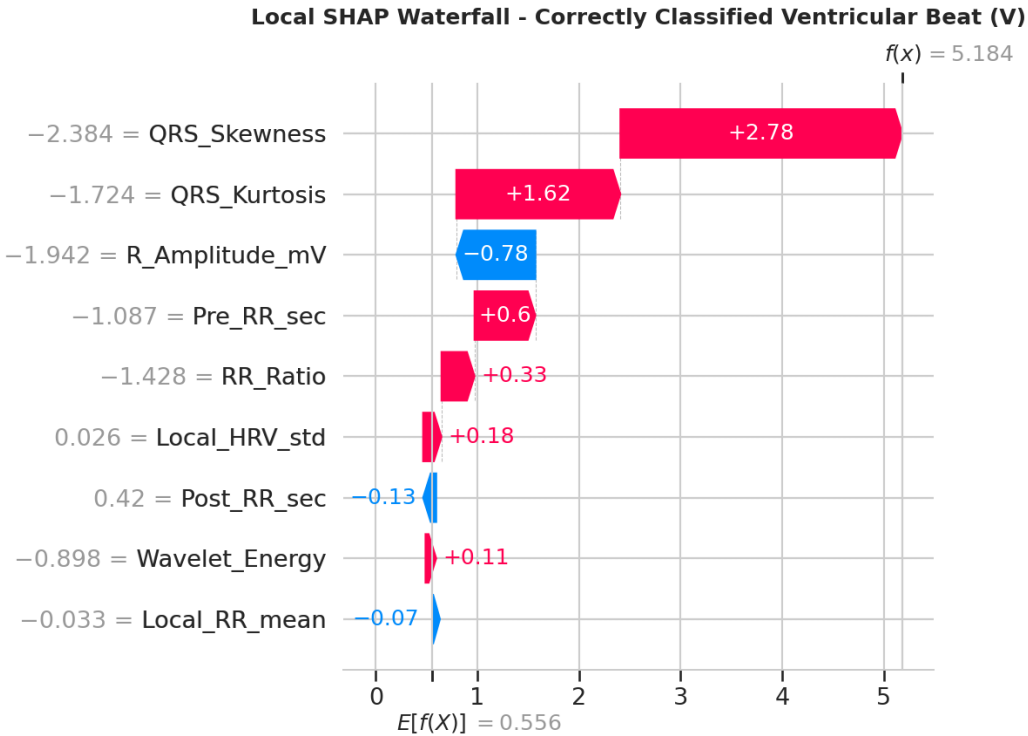


Interpretation

- Method: SHAP and Permutation Importance
- These features agree with clinical knowledge because arrhythmias often alter both heartbeat timing and waveform morphology
- Global explanations support the observed model performance

EXPLAINABILITY

Local Explainability: Why This Specific Prediction?



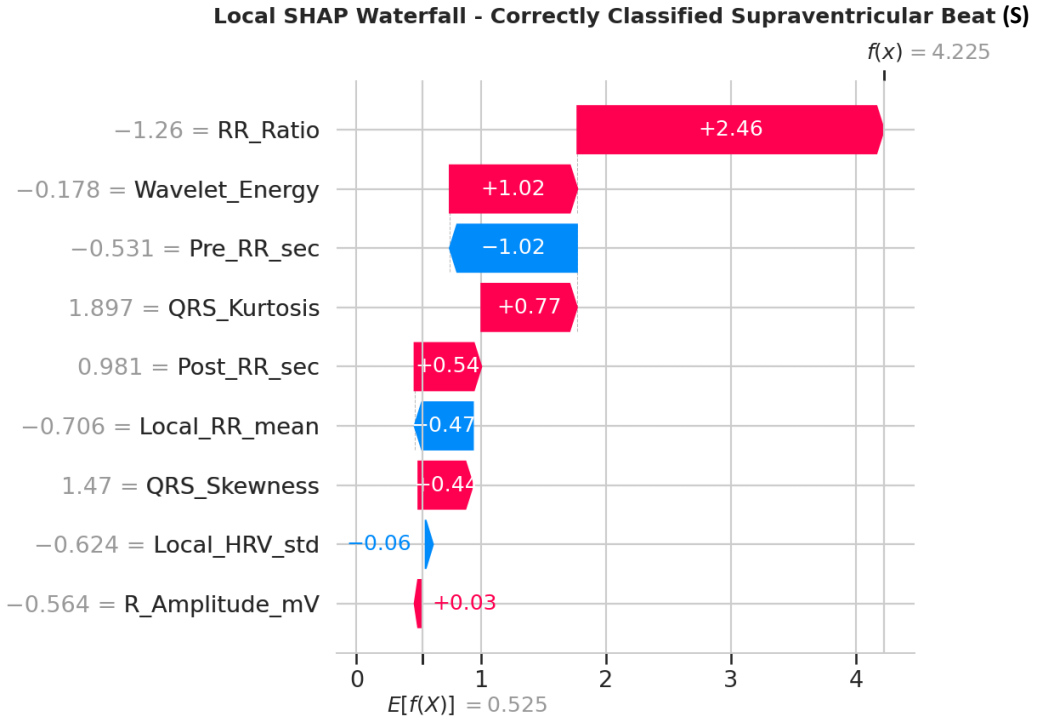
Case Analysis

Correct prediction:

- Kurtosis and skewness pushed the prediction toward Class V
- Consistent with global feature importance

EXPLAINABILITY

Local Explainability: Why This Specific Prediction?



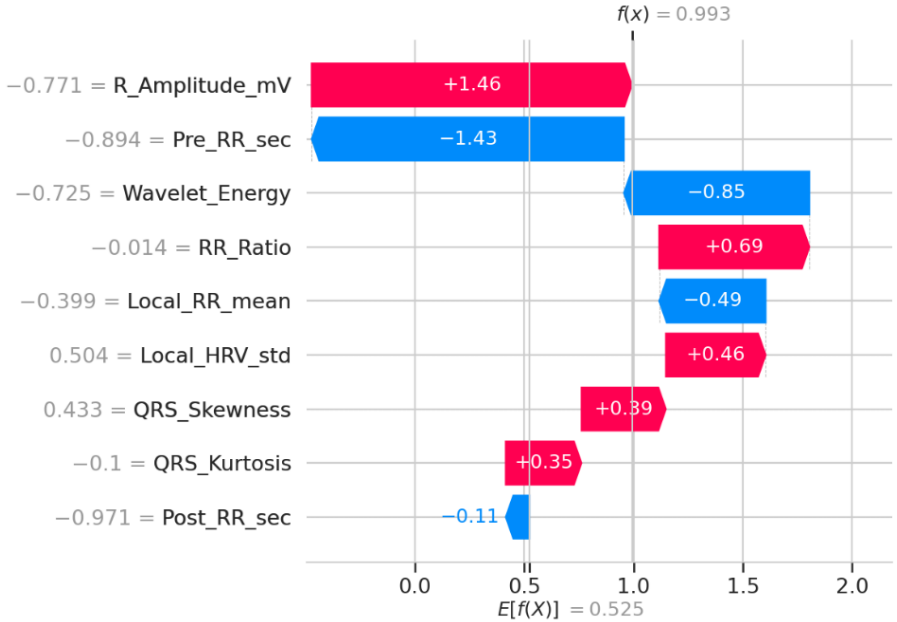
Case Analysis

Correct prediction:

- *RR_Ratio* and *Wavelet_Energy* heavily pushed the prediction toward the Supraventricular class (S)
- Rhythm-based features dominate the local decision, despite a strong opposing contribution from *Pre_RR_sec*

CRITICAL DISCUSSION

Error Analysis: What Did the Model Get Wrong and Why?



Reverse-Engineering Errors

- Most errors occur when Class S beats are classified as Normal
- 471 of 480 Class S errors follows this pattern

Misclassified prediction

- Elevated local heart rate masked the expected prematurity pattern.
- Temporal features become less informative
- The beat appeared similar to a normal heartbeat

Comparison with Prior Work

Similar difficulties are reported in prior ECG classification studies. Supraventricular beats remain one of the most challenging arrhythmia.

CRITICAL DISCUSSION

Domain Impact, Limitations & Future Directions

Domain Impact

- Assist cardiologists during ECG screening
- Reduce manual workload
- Support early arrhythmia detection

Limitations

- Severe class imbalance
- Small number of handcrafted features
- Static 200 ms extraction window
- Evaluation performed on a single dataset

Future Directions

- Dynamic heartbeat windows
- Hybrid deep-learning segmentation
- Additional physiological features
- Validation on external datasets

Thank You

Conclusions:

- We developed a machine learning pipeline for ECG arrhythmia classification
- A weighted soft-voting ensemble achieved the best overall performance under inter-patient validation
- Explainability analysis confirmed clinically meaningful decision patterns
- Future work will focus on adaptive feature extraction and external validation

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